

Evaluating the efficacy of AI systems in diabetic retinopathy detection: A comparative analysis of Mona DR and IDx-DR

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Abstract

Purpose: To compare two artificial intelligence (AI)-based Automated Diabetic Retinopathy Image Assessment (ARIA) softwares in terms of concordance with specialist human graders and referable diabetic retinopathy (DR) diagnostic capacity.

Methods: Retrospective comparative study including 750 consecutive diabetes mellitus patients imaged for non-mydratic fundus photographs. For each patient four images (45 degrees field of view) were captured, centered on the optic disc and macula. Images were manually graded for severity of DR as no DR, any DR (mild non-proliferative diabetic retinopathy [NPDR] or more), referable DR (RDR (more than mild DR)), or sight-threatening DR (severe NPDR or more severe disease and/or clinically significant diabetic macular edema [CSDME]). IDx-DR and MONA DR output was compared with manual grading and with each other.

Results: Total sample size was 750 patients, of which 55 were excluded due to ungradable images. Out of the remaining 695 patients 522 (75%) were considered as having no DR by manual consensus grading, and 106 (15%) as having RDR. Agreement between raters varied between moderate to substantial. IDx-DR showed moderate agreement with human grading ($k=0.4285$) while MONA DR had substantial agreement ($k=0.6797$). Out of 106 patients with a ground truth of RDR, IDx-DR identified 105 and MONA DR identified 99. The sensitivity and specificity rates for RDR detection of IDx-DR were 99.1 and 71.5% compared with MONA DR of 93.4 and 89.3% respectively. Of note, both ARIAs had 100% sensitivity for the detection of STDR.

Conclusion: Both ARIAs performed well in this study population, both with sensitivity for RDR screening over 90%, with IDx-DR showing higher sensitivity and MONA DR higher specificity. MONA DR showed superior agreement with human certified graders.

KEYWORDS

artificial intelligence, diabetic retinopathy, machine learning, ophthalmology

1 | INTRODUCTION

Diabetic retinopathy (DR) poses a significant global health burden. The escalating prevalence of diabetes worldwide has led to an alarming rise in DR, further reinforcing the need to establish efficient and cost-effective screening. Despite public health efforts, diabetes and diabetic-complication rates continue to surge, demonstrating a pressing need for cost and labor effective screening strategies (Grzybowski et al., 2020).

Traditional DR screening methods based on manual grading have been challenging to implement globally due to their resource-intensive nature, hindering diagnosis and management of sight-threatening DR (Grzybowski et al., 2020; Ting et al., 2016). DR screening programs have shown to have varying sensitivities, require time and the expertise from experienced specialists. To address these challenges, researchers and clinicians have turned to technology-driven, artificial intelligence-based solutions, such as Automated DR

Image Assessment software (ARIA). ARIAs present a new avenue for revolutionizing DR screening by leveraging image analysis and risk stratification algorithms to automatically grade fundus pictures, at the same time promising scalability and cost-effectiveness compared to manual grading (Grzybowski et al., 2020).

The development and deployment of ARIA systems have gained momentum, resulting in the availability of various ARIA solutions worldwide, with ongoing efforts to refine and enhance their capabilities. However, despite the growing number of ARIA solutions and implementations, direct comparisons between different ARIA platforms remain scarce in the scientific literature, which might hinder stakeholders from making decisions on whether a significant change in clinical practice is warranted, due to a lack of a clear direction. Existing research has demonstrated considerable variation in the performance of modern algorithms, highlighting the need for comprehensive evaluation and benchmarking studies (Grzybowski & Brona, 2021; Gunasekaran et al., 2020; Lee et al., 2021; Looker et al., 2014; Nielsen et al., 2019).

In light of this critical knowledge gap, the present study aims to evaluate and compare the performance of two contemporary DR screening solutions, IDx-DR and MONA DR, with consensus human expert grading.

2 | METHODS

2.1 | Retinal image acquisition

All images were taken from a DR screening initiative based in Greater Poland voivodeship in Poland. The images were taken in diabetic clinics and screening was offered to patients visiting for routine appointments related to their diabetes care. All patients signed an informed consent form to participate in this study, which was approved by the local institutional review board (approval number NR 11/2024). The screening process involved non-mydriatic fundus images captured using a Topcon NW-400 camera by trained operators who followed a specific imaging protocol. For each patient, a total of four images (45 degrees field of view each) were captured. They included one image centered on the optic disc and one centered on the macula for each eye. Additional images with the same centration were taken to ensure sufficient quality, when prompted by the IDx-DR image submission software. Retinal images were obtained from 750 consecutive patients with established diabetes mellitus and no previous DR diagnosis who underwent successful screening for DR using IDx-DR. Patients with images deemed of sufficient quality by the integrated IDx-DR quality assurance module were selected. A total of 3482 sufficient quality images from 750 patients were used for the study.

2.2 | Reference standard grading

The patients with images of sufficient quality as determined by the integrated IDx-DR quality assurance

module were split into two sub-datasets: 320 patients were each graded by three Polish retina specialists, 430 by three certified graders in India (two retinal specialists and a certified optometrist). Each eye of each patient was graded by three graders from either Polish or Indian group. At this stage graders were allowed to exclude patients from subsequent analysis if images were deemed ungradable.

All the graders were masked to the output of the AI and the output of the other graders. Images were graded for severity of DR based on International Clinical Diabetic Retinopathy (ICDR) severity classification as no DR, mild non-proliferative diabetic retinopathy (NPDR), moderate NPDR, severe NPDR or proliferative diabetic retinopathy (PDR) (Wilkinson et al., 2003). Macular edema was determined by the presence of surrogate markers like hard exudates. If hard exudates were found within 1 disc diameter (DD) of the fovea, macular edema was labelled as clinically significant diabetic macular edema (CSDME) present.

Image grading was done on a per eye basis. The final diagnosis for each patient was determined by the stage of DR in the more affected eye. Graders had access to all images taken for a given patient, including images with low quality, that were later retaken for the purpose of IDx-DR submission (as the IDx-DR quality assurance system suggests operators which images should be retaken). The majority image grading was regarded as the final reference standard and was based on Polish or Indian graders results, depending on which were available for a given patient. All statistical analyses were performed at the patient level.

2.3 | Differences in image analysis workflow

All the graders were masked to the output of the AI and the output of the other graders. Images were graded for severity of DR based on ICDR severity classification as no DR, mild non-NPDR, moderate NPDR, severe NPDR or PDR [10]. Macular edema was determined by the presence of surrogate markers like hard exudates. If hard exudates were found within 1 DD of the fovea, macular edema was labelled as CSDME present.

Image grading was done on a per eye basis, with each eye graded by three graders total. The final diagnosis for each patient was determined by the stage of DR in the more affected eye. Graders had access to all images taken for a given patient, including images with low quality, that were later retaken for the purpose of IDx-DR resubmission (as the IDx-DR quality assurance system suggests operators which images should be retaken). MONA DR, which uses a different image quality analysis, did not have this option and was fed the 3482 images collected during the screening, which were fed in bulk to the system in conventional image file formats. In practice, MONA DR requires only one image, centered on the macula, per eye, whereas IDx-DR requires one image centered on the optic disc and one centered on the macula. Majority decision was regarded as the final reference standard and was based on Polish or Indian graders results, depending on which were available for

a given patient. We determined the agreement of grades between groups of graders using (Fless et al., 2003; Dow et al., 2023) kappa coefficient, which is used to assess the concordance of more than two graders under the assumptions that the class sizes in the analysed sample are unequal. All statistical analyses were performed at the patient level. The results obtained for IDx-DR are from use in a standard screening workflow. This means that IDx-DR quality assurance module was able to instruct the operator to retake one or more images that it judged to be of insufficient quality and use those new images for screening. Only the four best images (per patient) were analysed by IDx-DR. Due to the way the images were anonymized and the absence of a human in the loop, it was not possible to only feed the required images into MONA DR. All images deemed to be of sufficient quality by MONA DR were analysed and the maximum value was used to obtain the patient-level result.

2.4 | AI-algorithms

MONA DR DME (Version 1.0.0; MONA. health, Bertem, Belgium) primarily consists of two ensembles, one for DR and one for DME. The DR ensemble consists of six unique regression models, with the result of two of them having twice the weight in the final averaging operation. The models were trained using a MSE loss function. The DME ensemble consists of seven unique classification model, trained with a binary cross-entropy loss function, also averaged to produce a single output. All models were built out of one of following architectures: ResNet50, InceptionV3, Xception, EfficientNet B3/B5, DenseNet with a custom head (Global Average Pooling, Dense and Dropout layers) trained with random image augmentations (rotation, flipping, shift, image brightness/contrast alterations), a batch size of 16, an initial learning rate of 0.0001, automatic learning rate decay, transfer learning from ImageNet, and using regularisations (L1/L2 and dropout). The ensembles were thresholded to select a fixed operating point. Data was obtained from the EyePACS telemedicine platform and includes patient visits from screening centers in the USA. DR labelling was consistent with the ICDR, where DME labelling was based on the presence of hard exudates within one DD of the macula/fovea (Peeters et al., 2023; Wilkinson et al., 2003). The training and test splits were performed randomly according to a fixed percentage (64/20/16), resulting in a training split of 195 281 images for DR and 126 397 images for DME. A validation split of 41 468 images for DR and 31 318 images for DME and finally a test split of 16 772 patients for DR and 16 833 patients for DME were performed. The split was performed at the patient level to ensure that there was no overlap of patients between datasets. Missing or corrupted images for certain patients resulted in a split that deviated slightly from the initial percentages. Detailed metadata and testing/verification on the test population can be found in Peeters et al. (2023). Both training and validation data have very similar characteristics due to the random nature of the split. The

technical details of the IDx-DR model have been described elsewhere (Abramoff et al., 2018).

2.5 | Statistical analysis

All data was stored in Microsoft Excel and Comma Separated Values (CSV) files and was analysed using Python programming language along with Numpy, Pandas, Scikit learn and SciPy libraries. The diagnosis of the AI using MONA DR and IDx-DR AI systems were tabulated against the majority image diagnosis (reference standard) by constructing 2×2 table. The sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV) with 95% CIs were calculated. Inter-rater agreement for Polish and Indian graders was measured by calculating the Cohen's kappa coefficient (Cohen, 1960) which is a standard method in measuring the consistency of two graders' decisions for qualitative items. Cohen's kappa coefficient determines the resulting values in the range of −1 to 1, where values close to 1 indicate high decision conformity, 0—random decision conformity and −1—high decision inconsistency. In order to additionally evaluate the measure of agreement of the decisions of the analysed systems with the gold standard, Cohen's kappa coefficient was also used. The McNemar test is a non-parametric statistical test for comparing two dependent samples, in which the variables are nominal or binary (McNemar, 1947). This test, for example, can be used to look for differences between groups under experimental conditions. In machine learning, it is widely used in evaluating statistical differences between classifiers. (Bostanci & Bostanci, 2012). The McNemar test (with an alpha of 0.05) was applied to assess the statistical significance of differences between classifiers (McNemar, 1947). The tests were performed for scores on a 4-class scale and on a binary scale showing the occurrence of:

- No DR
- Any DR—defined as mild DR or higher
- Referable DR (RDR)—defined as moderate DR or higher
- Sight-threatening DR (STDR)—defined as severe NPDR or higher or signs of CSDME

3 | RESULTS

The study included the images of 750 patients. Patients that had screening encounters deemed of insufficient quality by IDx-DR were excluded prior to patient selection. Full demographic data were available for 274 patients with an average age of 57 years old (oldest 83, youngest 18 years old). There was almost an even split between sexes with 344 female and 244 male patients.

Information regarding race was not gathered; however it is safe to assume that the vast majority of patients were white caucasian due to local demographics. Fifty-five patients had at least one image marked as ungradable by MONA DR system and were excluded. An additional two patients were removed from STDR analysis as they

did not have a majority for a STDR diagnosis despite being labelled as RDR by consensus. Image quality was further evaluated as part of the clinical workflow of using the MONA DR system. **Figure 1** presents the STARD diagram of retinal image selection in the study.

Grades from the Polish and Indian graders were converted to No DR, any DR, RDR and STDR before computing consensus, using the aforementioned definitions. Based on majority grading, there was no evidence of DR in 522 patients (75%). Any DR was seen in 173 (25%), of which 106 (15%) were referable DR, and 24 (3%) sight-threatening DR. Agreement scores for ICDR grading for both groups of graders are shown in **Table 1**. This show moderate to substantial agreement rates for Indian and Polish grader groups according to the Landis-Koch score (Landis & Koch, 1977).

MONA DR gives a binary output (RDR or no RDR) based on a continuous score between 0–4 and a threshold for RDR which has been reported previously. The analyses in this study were performed using this binary label. It detected the presence of RDR in 162 patients (23.3%). It correctly identified 99 of the 106 patients with a ground truth diagnosis of RDR. This translates to a

sensitivity of 93.4% (95% CI: 86.7%, 97.3%) and a specificity of 89.3% (95% CI: 86.5%, 91.7%). IDx-DR AI system gives an output at a stage level. 459 patients (66.0%) were flagged positive for DR (mild, moderate or sight-threatening). 273 of them were also categorized as RDR (39.3%, moderate or sight-threatening) and 112 as STDR (16.1%). Out of 106 patients with ground truth labelled as RDR, the IDx-DR system detected 105. This translates to a sensitivity of 99.1% (95% CI 94.7%, 100.0%) and a specificity of 71.5% (95% CI 67.6%, 75.1%). The diagnostic ability of both the AI systems including sensitivity, specificity, positive and NPVs are tabulated in **Tables 2** and **3**. It should be noted that, although the performance of MONA DR has been assessed against the ‘any DR’ label in this study, it is outside its intended use since the

TABLE 1 Values of Fleiss' kappa coefficients depicting agreement between graders.

	Fleiss' kappa
Polish	0.44
Indian	0.54

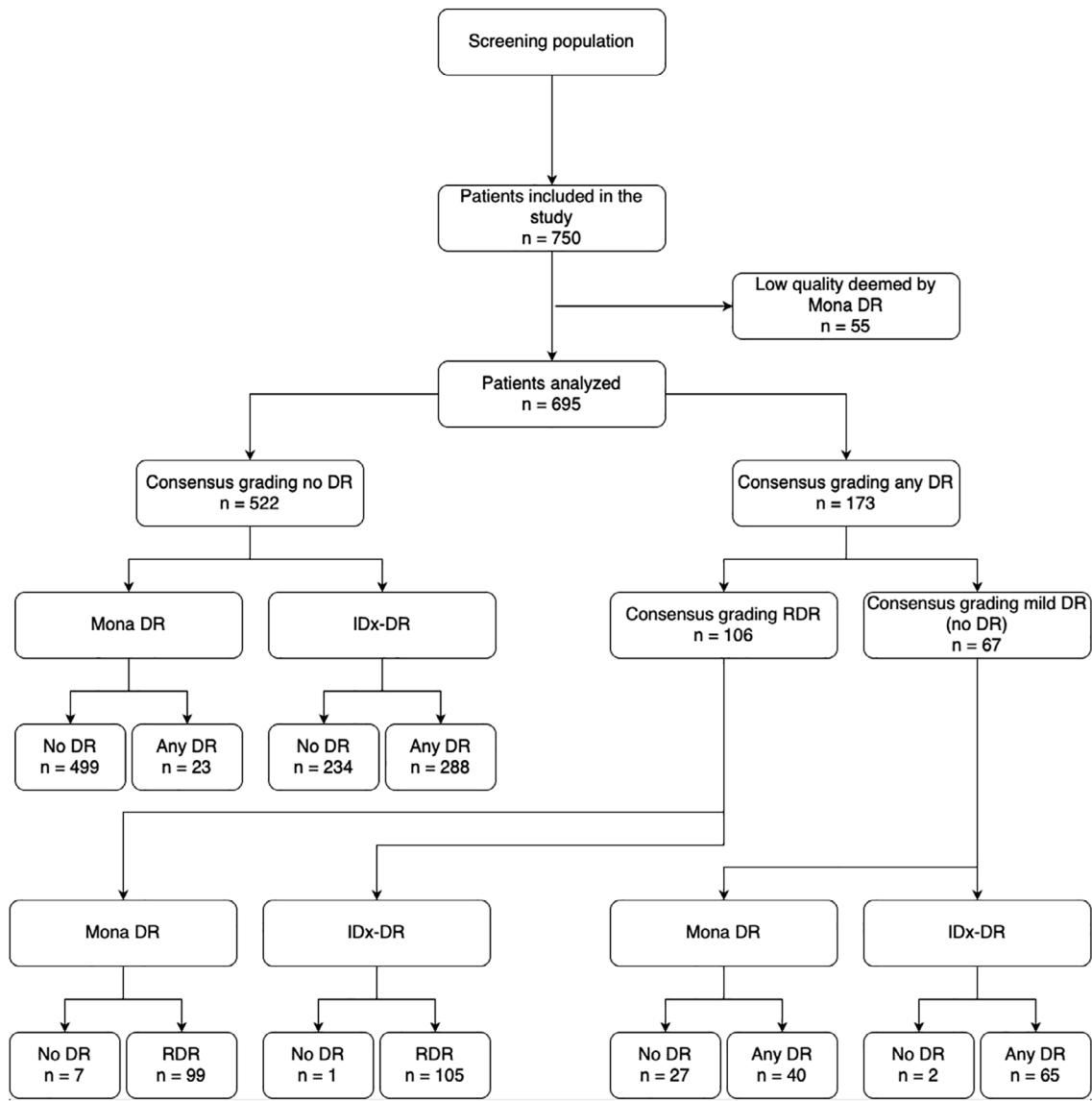


FIGURE 1 STARD diagram of retinal image selection in the study.

TABLE 2 Performance of analysed systems compared with consensus grading cutoff for DR detection at various levels.

			Automated diabetic retinopathy image assessment				Total
			MONA DR		IDx-DR		
			Positive	Negative	Positive	Negative	
Reference standard (consensus grading)	Any DR	Positive	139 (20%)	34 (5%)	171 (25%)	2 (0.2%)	173
		Negative	23 (3%)	499 (72%)	288 (41%)	234 (33%)	522
		TOTAL	162	533	459	236	695
	RDR	Positive	99 (14%)	7 (1%)	105 (15%)	1 (0.1%)	106
		Negative	63 (9%)	526 (76%)	168 (24%)	421 (60%)	589
		TOTAL	162	533	273	422	695
	STDR	Positive	24 (3%)	0 (0%)	24 (3%)	0 (0%)	24
		Negative	138 (20%)	533 (77%)	88 (13%)	583 (84%)	671
		TOTAL	162	533	112	583	695

Abbreviations: DR, diabetic retinopathy; RDR, referable diabetic retinopathy; STDR, sight-threatening diabetic retinopathy.

TABLE 3 Performance analysis of AI systems with 95% CIs compared to reference standard. Sensitivity, specificity and predictive values for both ARIAs were calculated against the different DR labels.

	Any DR		RDR		STDR	
	MONA DR	IDx-DR	MONA DR	IDx-DR	MONA DR	IDx-DR
Sensitivity (95% CI)	80.4 (73.6, 86.0)	98.8 (95.9, 99.9)	93.4 (86.9, 97.3)	99.1 (94.9, 100.0)	100.0 (85.8, 100.0)	100.0 (85.8, 100.0)
Specificity (95% CI)	95.6 (93.5, 97.2)	44.8 (40.5, 49.2)	89.3 (86.5, 91.7)	71.5 (67.6, 75.1)	79.4 (76.2, 82.4)	86.9 (84.1, 89.4)
PPV (95% CI)	86.9 (79.5, 90.8)	37.4 (32.8, 41.9)	60.3 (53.2, 68.7)	38.1 (32.7, 44.5)	14.6 (9.7, 21.2)	21.6 (14.2, 30.2)
NPV (95% CI)	93.5 (91.2, 95.5)	99.3 (97.0, 100.0)	98.6 (97.3, 99.5)	99.8 (98.7, 100.0)	100.0 (99.3, 100.0)	100.0 (99.4, 100.0)
F1 score (95% CI)	82.8 (78.5, 86.8)	54.1 (50.1, 58.0)	73.7 (68.2, 79.0)	55.4 (50.2, 60.5)	25.8 (19.7, 32.7)	35.3 (27.3, 44.0)

Bold values represent calculated values for sensitivity, specificity, PPV and NPV, with the values in parenthesis being the 95% confidence interval.

Abbreviations: DR, diabetic retinopathy; NPV, negative predictive value; PPV, positive predictive value; RDR, referable diabetic retinopathy; STDR, sight-threatening diabetic retinopathy.

output is a binary label 'RDR' or 'no RDR'. Testing the RDR label against 'any DR' ground truth can lead to excessive 'false negatives' in patients with very mild DR that do not meet the criteria for RDR. In analogy, its sensitivity to detect STDR was determined using the RDR label (since patients with ground truth label 'STDR' should be detected by MONA DR as 'RDR'). However, the specificity needs to be interpreted with caution, since patients with moderate DR would be labelled as 'RDR' by MONA DR and considered as a false positive for STDR by the analysis, leading to an underestimation of its specificity (Table 3).

Cohen's Kappa coefficient for RDR recognition concordance between the MONA DR system and the gold standard was 0.68 (substantial agreement), while for IDx-DR was 0.43 (moderate agreement). McNemar's test showed that the two classifiers were statistically significantly different on each decision scale analysed.

4 | DISCUSSION

The integration of AI in medical imaging, particularly in the detection and grading of DR, represents a significant

stride in diagnostic methodologies and medical care. There currently is a considerable number of commercially available DR screening algorithms, some of which are FDA approved and others are CE marked in Europe (Rajesh et al., 2023). However, while most have shown high sensitivity rates (comparable or superior to human graders), diagnostic performance has varied substantially and specificity is often lower than human graders, which increases referrals for human contact (Grzybowski et al., 2020; Gunasekaran et al., 2020; Lee et al., 2021; Rajesh et al., 2023). Reported results are difficult to compare due to differences in DL model architecture, validation and testing settings, population imbalances, among other factors (Gunasekaran et al., 2020; Tsiknakis et al., 2021). Therefore, there is a call for head-to-head validation of ARIA softwares (Rajesh et al., 2023). A recent large comparative study using seven masked algorithms yielded varying results, underlying the need for further investigation and robust pre-implementation assessment (Lee et al., 2021; Zhelev et al., 2023).

In the present investigation, we aimed to directly compare two AI algorithms, IDx-DR and MONA DR, in the context of DR screening. IDx-DR is based on a multi-layered convolutional neural network (CNN) and was

the first fully autonomous AI-based screening software to be approved by the FDA, thanks to its pivotal study showing a sensitivity of 87.2% and specificity of 90.7% for referable DR (Abramoff et al., 2018). Another FDA approved DR screening software is EyeArt, which in its pivotal study showed sensitivity of 95.5% and specificity of 85% for detection of RDR (Ipp et al., 2021). Both these systems can be considered as reference standards for AI-based DR screening and are currently well established as population-based DR referral tools (Bhaskaranand et al., 2019; van der Heijden et al., 2018). MONA DR is also based on a CNN and includes an image quality assessment algorithm as well as an additional image processing algorithm (Peeters et al., 2023). MONA DR system has been studied and tested on several fundus photograph datasets, having previously shown an AUC over 97% for DR detection, and sensitivity over 99% for severe DR (severe NPDR or PDR) diagnosis, and overall high accuracy across different study populations and camera models (Peeters et al., 2023).

In our study, a dataset comprising retinal images from 750 patients diagnosed with diabetes mellitus was used for algorithm testing. Initially, these images underwent evaluation by a group of medical experts (three retinal specialists for the Polish graders and two retinal specialists and a certified optometrist for the Indian graders), who graded ocular pathology attributable to diabetic complications based on ICDR classification (Wilkinson et al., 2003). Subsequently, the same retinal images were subjected to analysis by the two ARIAs. The primary objective was to assess the concordance of AI-based grading with that of the expert evaluations. MONA DR system showed a higher agreement (substantial according to the Landis-Koch score) with the expert graders' consensus for RDR recognition compared to moderate agreement for IDx-DR (Cohen's Kappa coefficient 0.68 vs. 0.43) (Landis & Koch, 1977). This may suggest that the MONA AI system's grading is more aligned with the current clinical understanding and grading standards of DR. The cause for the discrepancy in agreement of both algorithms was not investigated in the current study but several factors may have contributed to the lower Kappa in the case of IDx-DR, namely lower specificity or image quality. However, this result from IDx-DR is in line with previous publications showing moderate agreement with human graders (Grzybowski et al., 2022; Sedova et al., 2022).

Regarding diagnostic performance, the outcomes of this comparative study revealed a high degree of accuracy in both ARIAs, albeit with apparent discernible disparities in their decision-making processes. IDx-DR showed sensitivity and specificity rates of 99.1% and 71.5% for the detection of RDR, while MONA DR showed 93.4% and 89.3% rates, respectively. Lee et al., published a study with seven AI DR screening algorithms which were compared head to head and showed a wide variation of sensitivity (50.98%–85.90%) and specificity (60.42%–83.69%) (Lee et al., 2021). This suggests both IDx-DR and MONA DR are comparably better than most available algorithms concerning these diagnostic metrics. While IDx-DR demonstrated a higher sensitivity in detecting severe forms (referable) of DR compared to the MONA AI system (99.1% vs. 93.4%), the latter system showed higher specificity for “any DR”

and referable disease, comparable with previous assessment of MONA DR and similarly to another head-to-head study recently published with IDx-DR versus Medios AI (Grzybowski et al., 2023; Peeters et al., 2023). This, along with MONA DR higher grader agreement, suggests its potential not only as a performant screening software but also as a valuable tool in specialist settings, where accurate and detailed grading of DR is critical. However, as the current study was performed in a screening context, assumptions about the performance in a specialist setting need to be made with caution and would need to be confirmed in this specific context. Moreover, while the detection of RDR was the main objective of detection by both algorithms and the analysis is heavily focused on comparing the ability of both systems and human certified graders in detecting this feature, we find inclusion of the any DR and STDR categories is clinically relevant. The detection of STDR by the algorithms, as the most severe category, is an important metric because a failed detection (false negative), even if numerically low, can have potentially disastrous consequences. Our study shows both MONA DR and IDx-DR have 100% sensitivity for this feature. On the other hand, the inclusion of the any DR category can help clinicians and software developers gauge the limitations of these algorithms when faced with the task of detecting mild forms of DR.

MONA DR could be instrumental in assisting clinicians (particularly those with less experience) and non-medical care takers involved in DR screening, in accurately grading DR, thus facilitating access to expert-level diagnostics. In fact, research showed that a fully autonomous IDx-DR offers higher sensitivity but lower specificity for more than mild DR (RDR) compared with retina specialists, which increases the number of false positives and number of referrals. However, when a human is introduced into a hybrid AI-human workflow in which retina specialists would overread a IDx-DR AI-referred case, researchers were able to improve specificity considerably.[25] In contrast, our results suggest that MONA DR, having shown in our study to have higher specificity, could conceivably be applied to such a workflow model, albeit with less need (or at least a reduced workload) for an additional human evaluator to check for false positive. Of note, both AI systems correctly detected all cases with STDR.

The dataset used in this study was collected in the context of a clinical workflow designed for the use of the IDx-DR AI system, meaning that one optic disc centered and one macular-centered image of sufficient quality according to AI system were collected per patient and fed to the IDx-DR algorithm. Although the MONA DR system is designed to work with a single macula-centered image, for this evaluation it was fed with all available images in the dataset (including the optic disc centered images). Despite the fact that MONA DR was trained on macula-centered images, the algorithm is spatially invariant, so the inclusion of optic disc centered images for MONA DR evaluation still allowed for DR detection with high accuracy. Although the inclusion of optic disc-centered images could, in theory, affect the algorithm's performance, the performance metrics we show suggest it did not, at least not in a significant manner.

While these results are promising, there are limitations to this study. The AI systems were tested on a specific cohort of patients, and their performance might vary in different demographic and clinical settings. Exploring the AI systems' performance across diverse patient populations would be valuable in assessing their generalizability and utility in global health settings. There are a number of regional differences in DR screening model workflows, from fully autonomous to AI referral and human overread, and it is still unclear which would be the most cost-effective (Bellemo et al., 2019). Efforts should be done to compare the diagnostic utility (performance, cost-effectiveness, cost-utility) on a wider range of clinical settings, from the primary care, hospitals of differing magnitude and against different standards of grading, such as those performed exclusively by retina specialists but also by general ophthalmologist and optometrists." Additionally, the study focused on the comparison of two AI systems; a broader analysis involving multiple AI systems could provide a more comprehensive understanding of the capabilities and limitations of current AI technologies in DR screening. Future research should focus on the longitudinal analysis of AI system performance, wider algorithm comparisons, validation against differing population sets, ARIA integration into different clinical workflows and patient settings, and the impact on patient outcomes, both in terms of clinical outcomes and vision-related quality of life.

Furthermore, the continuous development of already commercially available screening platforms should be encouraged considering recent advances in AI-based DR diagnostics, to further tackle common problems in AI (such as generalizability and explainability) and improve overall performance and clinical applicability. For instance, DeepDR (Deep-learning DR) is a recently divulged and externally validated DL model which showed high sensitivity even for mild NPDR, with specificity increasing with severity. This DL model used a lesion-aware sub-network (as well as a quality assessment sub-network) to improve performance and explainability (Dai et al., 2021). The use of transformers (DL architectures different to those usually employed in CNN models) might also be helpful in improving model accuracy, generalizability and explainability (Bala et al., 2023). Augmenting CNN models with fuzzy clustering methods and attention modules may improve model accuracy while reducing computing needs and model size (Farag et al., 2022; Naz et al., 2022). Finally, there is recent evidence that DL models may be used to predict development or worsening of DR, which would further expand the interest, applicability and value of these technologies (Bora et al., 2021; Bora et al., 2023).

In summary, both AI systems demonstrated excellent performance in grading retinal images for DR. The MONA DR system showed a remarkable alignment with the expert consensus, indicating its sophisticated algorithmic design that seems to closely mirror clinical judgement. As AI technology continues to evolve, AI-based DR detection systems are poised to play a pivotal role in advancing medical care, improving diagnostic accuracy, and ultimately, contributing to better patient outcomes in the realm of DR.

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CONFLICT OF INTEREST STATEMENT

S.R. is the chief technology officer, co-founder and shareholder of MONA. health. He contributed to the current work by providing the predictions and information on the model. I.S. is co-founder, shareholder, and consultant of Mona. health, a KU Leuven/VITO spin-off to which the described MONA DR DME model has been transferred. She is also a consultant, or received a research grant from Alcon, Aerie, Allergan/AbbVie, Bausch&Lomb, Elios Vision, EyeD, Horus, Omikron, Santen and Laboratoires Théa. J.J. is consultant for Janssen Pharmaceutica, Bayer, Mona. health and Roche. Under her terms of employment at KU Leuven, J.J. is entitled to stock options in Mona. health. F.P. is a consultant for Novartis, Bayer, Abbvie, Horus Pharma, Laboratoires Théa and MONA. health.

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